
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is a core reliability requirement for machine
2 learning systems deployed in open-world environments, where test inputs may not
3 match the training distribution. Recent work shows that unlabeled in-the-wild data
4 can improve OOD detection, but the quantitative benefit depends on separating
5 useful OOD evidence from mixtures that also contain in-distribution (InD) samples.
6 Without a clean OOD set, the detector must be trained from comparisons inside this
7 mixed unlabeled pool. We introduce Medix, a median-centric framework that iden-
8 tifies candidate outliers from unlabeled data and then trains an OOD classifier using
9 those candidates together with labeled InD data. The median operation provides a
10 robust reference for central gradient behavior, allowing deviations induced by OOD
11 samples to be measured stably despite contamination. Theoretically, we derive
12 error bounds showing that Medix controls both inlier and outlier filtering errors
13 under explicit concentration, contamination, and separation terms. Empirically,
14 Medix consistently improves over existing methods in open-world evaluations,
15 with comparisons across datasets, metrics, and baselines supporting the practical
16 significance of the theoretical guarantees.

17 1 Introduction

18 Real-world deployment exposes machine learning models to a measurable reliability gap: inputs can
19 arrive from out-of-distribution (OOD) categories that are absent from the training label space and
20 therefore should not receive ordinary in-distribution predictions. This gap is practically important
21 because neural networks can assign confident predictions to such inputs while lacking a built-in
22 mechanism for identifying and rejecting them (Nguyen et al., 2015).

23 The central difficulty is that OOD distributions are not observed as labeled negative classes during
24 ordinary training, so the detector must infer a boundary between InD and OOD behavior from
25 limited evidence. Recent methods (Katz-Samuels et al., 2022a; Du et al., 2024a) therefore use
26 additional “in-the-wild” data as empirical evidence for the deployment environment. In particular,
27 Katz-Samuels et al. (2022a) regularize training with unlabeled wild data while preserving the labeled
28 InD classification objective. This source of data is attractive because it can be collected after
29 deployment and can better match the OOD patterns encountered at test time, providing a more
30 favorable basis for quantitative OOD detection than InD-only training.

31 The same wild-data setting also creates a precise comparison problem: the unlabeled pool is a mixture
32 of InD and OOD samples, but the detector is not told which examples belong to which component. A
33 method must therefore extract an informative OOD subset without changing the empirical meaning
34 of the mixture. This setting remains underexplored, particularly from a theoretical perspective; to the
35 best of our knowledge, Du et al. (2024a) is the only prior work that provides formal guarantees for
36 the “in-the-wild” setting.

At the same time, median-based data pruning has shown that robust centrality can expose distributional deviations in contaminated data (Acharya et al., 2024). This evidence motivates the following question, framed around whether the same robustness can yield quantifiable OOD filtering guarantees:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question by introducing a median-centric perspective for OOD detection. Our algorithm, Medix, solves a median-based filtering problem that selects candidate OOD samples from wild data and then uses them to train a detector. The quantitative structure of the guarantee is explicit: the fraction of InD samples incorrectly flagged as outliers is bounded by a contamination effect, which remains controlled when the OOD proportion is below 50%, and a concentration effect, which uses the sub-Gaussian behavior of InD gradients to limit deviations from the InD mean gradient. These terms make clear when filtering errors remain small with high probability and give a theoretical explanation for the empirical margins reported below.

Our empirical comparison is organized around two baseline groups: methods trained only on InD data and methods trained with both InD and auxiliary unlabeled data. On CIFAR-100 (Krizhevsky et al., 2009), Medix improves average FPR95 over the strong KNN+ baseline (Sun et al., 2022) by 40.98%, showing the practical value of the wild-data signal. Unlike Outlier Exposure (Hendrycks et al., 2019), which assumes a clean auxiliary OOD set separable from InD data, Medix obtains these gains from mixed wild data. Against WOODS (Katz-Samuels et al., 2022a), the closest wild-data baseline, Medix further reduces average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10.

Our key contributions are as follows:

- C1)** We propose Medix, a median-centric greedy filtering approach that first extracts candidate outliers from unlabeled wild data and then trains an OOD detector by comparing those candidates against labeled InD samples. The primary algorithmic contribution is the filtering stage.
- C2)** We establish quantitative guarantees for median-based filtering in unlabeled mixtures. The bounds separately track InD samples incorrectly flagged as outliers and OOD samples incorrectly retained as inliers, with the dominant terms decomposed into contamination effects, which remain controlled below 50% OOD proportion, and concentration effects, which stabilize gradient behavior.
- C3)** We evaluate Medix on eleven InD-OOD pairs against 20 competitive baselines. Across these comparisons, Medix achieves the strongest reported OOD detection performance, and the extraction experiment reports an outlier-filtering error rate of only 12.5% (Figure 2), aligning the empirical pattern with the theory.

2 Preliminaries and Problem Setup

This section fixes the comparison setting used throughout the paper: the labeled InD sample, the unlabeled wild mixture, the InD classifier, the OOD detector, and the metrics used to quantify detection and classification performance.

Labeled In-Distribution Data. Let $\mathcal{X}$ denote the input space and $\mathcal{Y} = \{1, \dots, K\}$ the InD label space. The labeled InD sample $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ consists of n i.i.d. pairs from the unknown joint distribution $\mathbb{P}_{X \times Y}$ over $\mathcal{X} \times \mathcal{Y}$. Its marginal on inputs is $\mathbb{P}_{\text{in}}$, and this sample is the empirical basis for training the InD model used in all later comparisons.

Out-of-distribution detection. We study the deployment condition in which training uses only labeled InD data, but inference may include inputs from labels outside $y \notin \mathcal{Y}$. The detector is evaluated by whether it separates these OOD inputs from samples drawn from $\mathbb{P}_{\text{in}}$, rather than by forcing every input into one of the known classes.

Unlabeled wild data. Because labeled OOD examples are expensive and incomplete relative to the size of the possible OOD space, we use unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, as the empirical source of OOD evidence, following Katz-Samuels et al. (2022a). The key evaluation condition is that this pool is not clean: it mixes InD and OOD samples collected during deployment of a model

trained on $\mathcal{S}_{\text{in}}$. As in Du et al. (2024a); Katz-Samuels et al. (2022a), we represent the mixture with the Huber contamination model (Huber, 1964):

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

where π is the OOD contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution over $\mathcal{X}$. The case $\pi = 0$ contains no OOD evidence and is therefore excluded as a trivial detection setting.

Models and Loss Functions. The InD classifier is $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$, with parameters $\phi \in \Phi$ and outputs over the $K = |\mathcal{Y}|$ InD classes. Its labeled-data loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For detection, we use a separate binary classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ with parameters $\theta \in \Theta$ and loss $\ell_b(g_\theta(x), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$. The sign convention is explicit: $y_+ > 0$ denotes InD and $y_- < 0$ denotes OOD.

Learning objective. The learning objective uses both $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$ to train the detector g_θ while retaining the multi-class classifier f_ϕ . At test time, the quantitative comparison is reported with the following metrics, where lower FPR and higher TPR/accuracy indicate better performance:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X}\mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

where $\mathbb{I}\{\cdot\}$ denotes the indicator function, and $\mathbb{P}_{\text{out}}^{\text{test}}$ represents the OOD test data distribution.

3 Method: Median-Centric Framework for OOD Detection

This section presents **Medix**, a two-stage framework that turns mixed unlabeled wild data into usable OOD training evidence. The first stage, **1) Outlier Extraction**, filters candidate OOD samples from the wild pool using the median-based criterion in Section 3.1. The second stage, **2) Detector Training**, trains a binary OOD detector from labeled InD samples and the extracted candidates, following the protocol of Du et al. (2024a) as described in Section 3.2. This separation makes the comparison structure clear: Medix improves detection by changing the quality of the OOD candidates supplied to the same downstream detector-training objective.

3.1 Extracting Candidate Outliers from the Wild Data

To isolate candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix compares gradient behavior in the wild pool against a reference estimated only from labeled InD data. The gradients come from the classifier f_ϕ , trained solely on $\mathcal{S}_{\text{in}}$, so deviations from the reference have a clear quantitative interpretation as departure from InD behavior.

Reference gradient estimation from InD data. The first step estimates the InD reference gradient. We train f_ϕ on $\mathcal{S}_{\text{in}}$ by empirical risk minimization (ERM), producing the baseline model used for all subsequent gradient comparisons:

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. After training, the InD baseline is summarized by the mean gradient $\bar{\nabla}_{\text{in}}$, computed over the labeled InD sample:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

This vector $\bar{\nabla}_{\text{in}}$ is the reference point against which Medix measures whether subsets of wild data behave like InD data or exhibit OOD-induced deviation.

Motivation. The empirical motivation is a controlled comparison in which the OOD proportion in $\mathcal{S}_{\text{wild}}$ is varied while the InD reference remains fixed. Using CIFAR-10 (Krizhevsky et al., 2009) as InD and SVHN (Netzer et al., 2011) as OOD, we start with 10,000 CIFAR-10 wild samples

disjoint from the training set for $\phi_{\mathcal{S}_{\text{in}}}$. We then incrementally add SVHN samples and measure the L_2 deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of wild-data gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

Figure 1 shows a clear monotonic increase in this deviation as additional SVHN OOD samples enter the wild set. This empirical pattern directly motivates Medix: the algorithm removes samples that most reduce the median-to-reference distance and stops when the improvement in consecutive L_2 deviations falls below the threshold ϵ .

Filtering potential outliers from unlabeled wild data. Motivated by the monotonic pattern in Figure 1, we formulate filtering as the problem of finding the wild subset whose EWM is closest to the InD reference, with the complement interpreted as the candidate outlier set $\mathcal{S}_{\text{out}}^*$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$. The objective selects the subset $\mathcal{S}$ whose gradient median has the smallest distance to $\bar{\nabla}_{\text{in}}$. In light of Figure 1, the minimizer $\mathcal{S}_{\text{in}}^*$ is expected to retain the InD portion of the wild mixture, while $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ provides the OOD candidates used for detector training.

Solving (4) exactly is computationally prohibitive for large $\mathcal{S}_{\text{wild}}$. Algorithm 1 therefore uses a leave-one-out greedy approximation that ranks samples by how much their removal improves the EWM distance to the InD reference.

At iteration t , the algorithm computes the EWM of gradients for the remaining wild set $\mathcal{S}$ and records $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. For each candidate removal $i \in \mathcal{S}$, it recomputes $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ and the corresponding distance drop δ_i . The k samples with the largest drops are treated as the strongest contributors to OOD deviation and moved to the estimated outlier set. The convergence rule follows the preliminary pattern in Figure 1: once many OOD samples have been removed, the distance reduction becomes small, so Medix stops when the improvement falls below ϵ or the maximum iteration count is reached. This stopping rule is designed to retain InD samples while extracting the OOD candidates that most improve the median alignment.

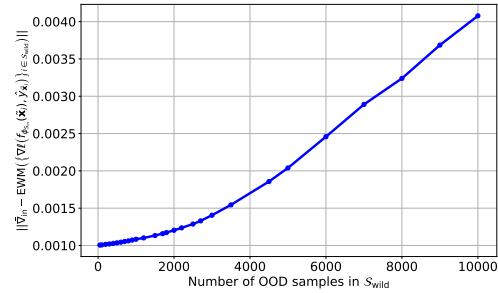


Figure 1: Median-to-InD gradient distance increases as SVHN OOD samples are added to the CIFAR-10 wild mixture $\mathcal{S}_{\text{wild}}$.

Algorithm 1 Iterative Outlier Detection via Medix

Require: $\bar{\nabla}_{\text{in}}$, $G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i}})$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

3.2 Training the OOD Detector with Candidate Outliers

After Medix extracts the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$, the detector-training stage compares those candidates directly against labeled InD samples. Following Du et al. (2024a), the binary detector g_θ is trained so that labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ receive positive scores and candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$ receive negative scores, with the decision boundary thresholded at 0. The loss is:

$$\begin{aligned}\mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}.\end{aligned}\tag{5}$$

Because the 0/1 loss is non-differentiable, optimization uses a differentiable sigmoid-based binary loss as a smooth surrogate while preserving the same InD-versus-outlier comparison. The training objective also includes the InD risk from Equation 2, so the detector gains OOD separation without discarding the InD classification signal. The empirical results below quantify the resulting generalization of g_θ to OOD samples from $\mathbb{P}_{\text{out}}$.

4 Theoretical Analysis

We now state the guarantees for the Medix filtering stage in terms of the same errors used to interpret candidate extraction: InD points incorrectly removed as outliers and OOD points incorrectly retained as inliers. The following theorems provide upper bounds for both rates, making the roles of sample concentration, contamination proportion, and OOD-InD separation explicit. Detailed proofs appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Theorem 4.1 decomposes the InD filtering error into two interpretable quantities. The concentration term decreases with larger InD sample size and captures finite-sample gradient fluctuations, while the contamination term measures the influence of OOD points on the EWM. The bound remains controlled when $\pi < 0.5$, matching the median intuition that fewer-than-half contaminating samples cannot dominate the center. The complementary question is how many true OOD samples are retained as InD; the next theorem bounds that error rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1-\pi}{2\pi}}_{\text{Contamination term}}.$$

Theorem 4.2 complements the InD guarantee by bounding the fraction of OOD points mistakenly retained as InD. Together, Theorems 4.1 and 4.2 give a two-sided accounting of Medix filtering errors. The rates are controlled by the following effects, each of which has a distinct empirical interpretation:

178 **Contamination Effect.** In Theorem 4.1, the $\pi/[2(1 - \pi)]$ term quantifies how OOD contamination
 179 can skew the EWM; the median criterion stays favorable as long as $\pi < 0.5$. Theorem 4.2 con-
 180 tains the reverse penalty $\frac{1-\pi}{2\pi}$, reflecting the harder comparison regime in which OOD points are
 181 underrepresented in the wild mixture.

182 **Concentration Effect.** Both bounds use sub-Gaussian gradient coordinates to convert sample size
 183 into error control. For InD data, concentration near $\bar{\nabla}_{\text{in}}$ keeps the median stable; for OOD data,
 184 concentration around μ_{out} quantifies the chance that an OOD sample resembles the InD reference.
 185 These finite-sample terms decay as $1/\sqrt{m}$ for the relevant sample count.

186 **Separation Effect.** This term appears only in Theorem 4.2 and measures how far the OOD mean
 187 gradient is from the InD mean gradient. The exponential term $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ improves as
 188 separation grows, so larger gradient margins make retained OOD errors less likely.

Theorems 4.1 and 4.2 jointly show that, with high probability, median filtering separates InD and OOD samples in unlabeled mixtures with bounded errors on both sides. InD removals are controlled by contamination and concentration terms, while OOD retention is controlled by separation, concentration, and reverse-contamination terms. This decomposition explains why Medix can improve detector training when the wild data provide consistent gradient evidence.

189
 190 *Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates.* The sub-Gaussian assumption
 191 supplies the concentration terms used in the bounds, and the empirical diagnostics support its
 192 plausibility for the gradients used by Medix. Figure 4a shows InD gradient values that are bell-shaped
 193 and concentrated near the mean, indicating light-tailed behavior. Figure 4b compares empirical
 194 gradient quantiles with Gaussian quantiles in a Q-Q plot (Wilk and Gnanadesikan, 1968); the close
 195 alignment with the 45-degree reference line provides additional evidence that the gradient distribution
 196 is well approximated by a Gaussian, and hence by a sub-Gaussian model. *We also provide a looser*
 197 *version of these bounds, which **removes the sub-Gaussian assumption** and holds under merely*
 198 *bounded second moments. This version is presented in Theorem C.3 (Appendix C.3). While the rates*
 199 *degrade, the core robustness guarantee of Medix still holds in this broader setting.*

200 5 Experiments

201 This section quantifies Medix across multiple InD-OOD dataset pairs and compares it with 20
 202 widely used baselines under matched evaluation metrics. The results are organized to make the
 203 method, baseline group, metric, InD dataset, OOD dataset, and wild-data condition explicit. All
 204 experiments are performed on hardware equipped with NVIDIA A100-SXM4-80GB GPUs, and code
 205 for reproducing the results will be released upon acceptance.

206 5.1 Models, Datasets, and Baselines

207 **Datasets.** We follow the protocol of Katz-Samuels et al. (2022a), which introduced learning OOD
 208 detectors from wild data, so the comparisons are aligned with prior work. CIFAR-10 and CIFAR-100
 209 serve as InD datasets ($\mathbb{P}_{\text{in}}$), and PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi
 210 et al., 2014), LSUN-RESIZE, and LSUN-C (Yu et al., 2015) serve as OOD test datasets ($\mathbb{P}_{\text{out}}$). For
 211 the wild-data condition ($\mathbb{P}_{\text{wild}}$), we mix a subset of InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) using the
 212 default contamination parameter $\pi = 0.5$. For example, when PLACES365 is the OOD test set, the
 213 corresponding wild mixture combines CIFAR and PLACES365 samples; the same construction is
 214 repeated for each OOD dataset and baseline. Each CIFAR InD dataset is split into two halves: 25,000
 215 images train $\phi_{S_{\text{in}}}$, and the remaining images contribute to S_{wild} . Gradients are computed with respect
 216 to penultimate-layer weights, which are informative for OOD detection (Huang et al., 2021a).

217 **Evaluation metrics.** We report three standard metrics for each method-dataset pair: (1) FPR95 $\downarrow$,
 218 the false positive rate on OOD samples when the InD true positive rate is fixed at 95%; (2) AUROC $\uparrow$,
 219 the area under the receiver operating characteristic curve; and (3) InD Acc $\uparrow$, the classification accuracy
 220 on InD samples.

Baselines. We group baselines by their available training evidence. The InD-only group uses $\mathbb{P}_{\text{in}}$ and includes Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all built on softmax cross-entropy models. We also include contrastive-loss methods, CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022), to compare against stronger InD-only representations. The wild-data group uses both InD and wild data. It includes Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize confidence or energy on outlier data, as well as WOODS (Katz-Samuels et al., 2022a), the closest prior method for unlabeled wild data through constrained optimization. We also include recent baselines CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024) for broader coverage.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ is a Wide ResNet (Zagoruyko, 2016) with 40 layers and width factor 2. It is trained with stochastic gradient descent using momentum 0.9, weight decay 0.0005, initial learning rate 0.1, cosine learning-rate decay, batch size 128, dropout 0.3, and 100 epochs. For Medix, ϵ is selected from $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and k from $\{4k, 7k, 10k, 20k\}$, with dataset size and OOD performance guiding the choice. The OOD classifier g_{θ} is initialized from $\phi_{\mathcal{S}_{\text{in}}}$ and adds a binary linear layer on penultimate-layer features. It is fine-tuned for 100 epochs with learning rate 0.001 using Equation (5); the binary loss is combined with the InD classification loss using weight 10 on the binary component, and all other training parameters match the InD classifier.

Table 1: CIFAR-10 InD comparison across OOD datasets. Each method is evaluated by FPR95 $\downarrow$, AUROC $\uparrow$, and InD ACC $\uparrow$; values are averaged over five runs, and best results are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
	Using $\mathbb{P}_{\text{in}}$ only												
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
	Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$												
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: CIFAR-100 InD comparison across OOD datasets. Each method is evaluated by FPR95 $\downarrow$, AUROC $\uparrow$, and InD ACC $\uparrow$; values are averaged over five runs, and best results are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
	Using $\mathbb{P}_{\text{in}}$ only												
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
	Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$												
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

5.3 Results

Table 2 reports the main CIFAR-100 comparison. The first empirical pattern is that methods using both InD and wild data generally improve over InD-only methods, confirming that the wild mixture

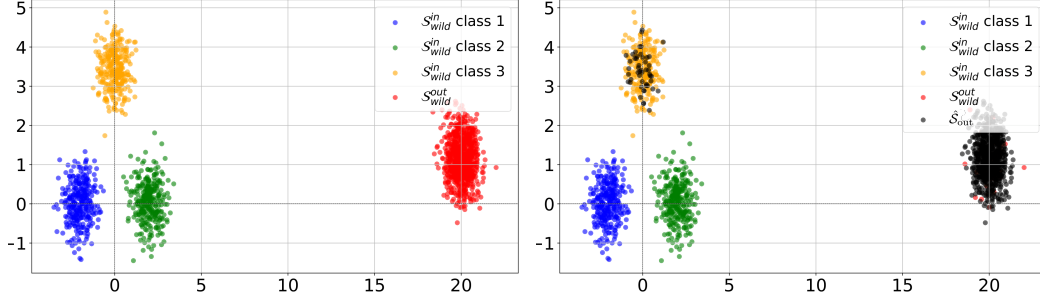


Figure 2: Medix filtering on a 2D wild mixture. (a) InD samples S_{wild}^{in} are drawn from three multivariate Gaussian classes and OOD samples S_{wild}^{out} from a separated Gaussian. (b) Medix-selected outliers $\hat{S}_{out}$ are shown in black; only 12.5% of the selected outlier set consists of InD samples.

contains useful OOD evidence. Relative to the strong InD-only KNN+ baseline, Medix lowers FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES. The second pattern is that Medix improves within the wild-data group: on CIFAR-100 it achieves an average FPR95 of 5.42%, a 1.32% improvement over WOODS, while maintaining 73.33% InD accuracy. This InD accuracy should be interpreted with the protocol in mind: Medix trains from 25,000 labeled InD samples, whereas InD-only baselines use the full 50,000-image CIFAR-100 training set. Table 1 shows the same comparison with CIFAR-10 as InD data. Medix again gives the best OOD detection results across the reported baselines, including FPR95 improvements over WOODS of 7.21% on PLACES365 and 5.25% on TEXTURES. The consistency across CIFAR-10 and CIFAR-100 indicates that the filtering advantage is not tied to a single InD dataset.

A representative visual example of Medix. We further isolate the extraction stage of Algorithm 1 on a 2D synthetic mixture where the ground-truth InD/OOD status is visible. InD samples come from three multivariate Gaussian classes with means $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$, all using covariance $0.25 \cdot I$. OOD samples come from $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. Figure 2 shows that the selected outliers largely match the ground truth: Medix flags 87.5% of actual OOD samples, equivalently leaving only a 12.5% InD contamination rate in the selected outlier set.

Additional studies and insights. Appendix A reports additional quantitative checks that stress different parts of the comparison. These include hyperparameter sensitivity for ϵ and k (Appendix A.2), EWM versus geometric median filtering (Appendix A.1), additional competitive contrastive and recent baselines (Appendix A.3), large-scale unseen OOD evaluation where $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4), computation and memory profiling (Appendix A.6), pseudo-label quality analysis (Appendix A.5), and comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

OOD detection has become a broad empirical benchmark area, with methods commonly compared by how well they rank InD and OOD inputs under metrics such as FPR95 and AUROC (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuricic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One major line of work defines post hoc scoring functions, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020). These methods provide important InD-only baselines in our tables.

Another line improves OOD behavior during training through regularization (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020;

283 Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b,
 284 2023; Wang et al., 2023a). Such methods reduce confidence (Lee et al., 2017; Hendrycks et al., 2019)
 285 or increase energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data, but the
 286 quantitative comparison often relies on access to an auxiliary OOD dataset. Medix targets the harder
 287 mixed-wild-data setting rather than assuming a clean auxiliary set.

288 Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the labeled-
 289 data-only assumption by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al.,
 290 2025), though they do not provide the same explicit median-based outlier extraction mechanism. Du
 291 et al. (2024a,b) do introduce filtering, but their thresholding approach differs from Medix, which
 292 uses median alignment of gradients. Moreover, Katz-Samuels et al. (2022a); Du et al. (2024a)
 293 assume batch-level mixing with controlled InD/OOD ratios in each batch, whereas large outsourced
 294 datasets are more naturally mixed at the dataset level. Medix is designed for this dataset-level mixture.
 295 Positive-unlabeled learning methods also train with positive and/or unlabeled data (Letouzey et al.,
 296 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020;
 297 Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022), but
 298 they primarily separate $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ rather than extracting outliers and training a dedicated OOD
 299 detector with provable filtering-error control.

300 7 Conclusions

301 We introduced Medix, a median-centric OOD detection framework that converts mixed unlabeled wild
 302 data into candidate OOD evidence for detector training. The theoretical analysis gives a quantitative
 303 account of both filtering errors: InD samples incorrectly removed as outliers are controlled by
 304 concentration and contamination terms, with robustness up to the 50% contamination threshold,
 305 and OOD samples incorrectly retained are controlled by separation, concentration, and reverse-
 306 contamination terms. The empirical comparisons show that these guarantees translate into practical
 307 gains: across diverse benchmarks and 20 baselines, Medix reduces average FPR95 by 40.98% relative
 308 to the strong KNN+ baseline, improves over wild-data methods such as WOODS and recent methods
 309 such as DRL, and achieves an outlier extraction error rate as low as 12.5%.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

We compare the element-wise median with the geometric median (Acharya et al., 2024) to justify which median statistic provides stronger OOD evidence for Medix filtering. The comparison uses CIFAR-10 as InD and SVHN as OOD. The wild set $\mathcal{S}_{\text{wild}}$ starts with 5k CIFAR-10 samples disjoint from the training set for $\phi_{\mathcal{S}_{\text{in}}}$, which provides $\bar{\nabla}_{\text{in}}$. We then add SVHN OOD samples and track both the L_2 -norm deviation from $\bar{\nabla}_{\text{in}}$ and the proportion of OOD samples removed. Figure 3 shows that, at the same OOD contamination level, the element-wise median identifies a higher proportion of OOD samples as outliers than the geometric median. This empirical margin indicates that EWM is more sensitive to the distributional shifts used by Medix, making it the more effective filtering statistic in this setting.

A.2 Hyperparameter Selection and Sensitivity Analysis

We evaluate sensitivity to the stopping threshold ϵ and removal budget k using CIFAR-100 as InD and SVHN as OOD. Table 3 shows that Medix remains strong across the tested settings. The best listed configuration, $\epsilon = 0.005$ and $k = 20000$, achieves FPR95 of 0.16. Reducing ϵ to 0.0005 yields FPR95 of 0.20, and halving k to 10000 yields FPR95 of 0.68. These values remain favorable relative to the listed baselines, showing gradual degradation rather than sensitivity to a single exact tuning choice.

A.3 Comprehensive comparison with recent competitive methods

We extend the comparison to competitive baselines with contrastive or recent learning objectives, including SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), CIDER Ming et al. (2023), CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c). This tests whether the Medix gains persist against stronger representation-learning baselines rather than only standard scoring methods. For this experiment, CIFAR-100 is the InD dataset and ResNet-34 is used as the model¹, following Ming et al. (2023); Zhang et al. (2024). Training uses stochastic gradient descent with momentum 0.9, weight decay 10^{-4} , 500 epochs, initial learning rate 0.5, cosine scheduling, and batch size 512. We use the checkpoints provided by Ming et al. (2023)². Table 4 shows that Medix achieves the best average FPR95 of 8.68% and average AUROC of 97.76%, while also providing filtering-error guarantees absent from most alternatives.

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

²<https://github.com>

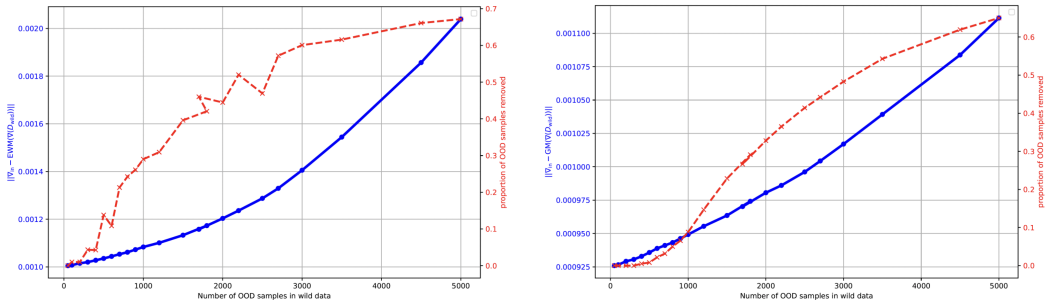


Figure 3: Element-wise median (EWM) versus geometric median (GM) under increasing OOD contamination, comparing how each statistic responds to distributional shift in the wild mixture.

Table 4: CIFAR-100 ResNet-34 comparison with recent and contrastive baselines. Results report FPR95↓ and AUROC↑ averaged over five runs; best values are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	±0.05	±0.02	±1.76	±0.36	±0.08	±0.03	±0.72	±0.24	±0.65	±0.16

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next test Medix in a harder unseen-OOD comparison. The setting increases difficulty in two ways: the wild OOD source $\mathbb{P}_{\text{out}}$ is large-scale, and the test-time OOD distribution $\mathbb{P}_{\text{out}}^{\text{test}}$ is deliberately different from the wild OOD distribution, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$. This separates memorization of the wild OOD source from practical generalization to unseen OOD data. CIFAR-100 is the labeled InD dataset, 300K Random Images Hendrycks et al. (2019) provides unlabeled wild OOD data, and SVHN is used only as the test OOD distribution. We compare against baselines that also use wild data during training. Table 5 shows that Medix obtains the lowest FPR95, 41.29 ± 1.2 , and the highest AUROC, 87.25 ± 0.6 , giving a large FPR95 margin over the wild-data baselines in this unseen-OOD condition.

Table 5: Large-scale unseen-OOD comparison, where CIFAR-100 is InD, 300K Random Images provide wild OOD data, and SVHN is the test OOD dataset.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

We also test whether pseudo-label confidence materially changes filtering quality. Using CIFAR-10 as InD and LSUN-Resize as OOD, we remove low-confidence pseudo-labeled samples with softmax probability below 0.6 before filtering. This pre-filter removes 15.98% of the samples from the training set.

The quantitative difference is negligible: both the filtered and unfiltered variants achieve FPR95 of 0.01%, while AUROC changes from 99.98% to 99.99% with filtering. This comparison indicates that Medix is already robust to noisy or low-confidence pseudo-labels, and that an additional confidence pre-filter provides little practical gain under this condition.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We profile the filtering stage to quantify its runtime and memory footprint. Experiments use an NVIDIA A100-SXM4-80GB GPU with approximately 15,000 unlabeled samples for each InD-OOD pair, using CIFAR-10 and CIFAR-100 as InD datasets and LSUN-Resize as OOD. Table 6 shows that peak GPU memory remains near 100 MB and current GPU memory near 32 MB, while wall-clock time is 4497.17 seconds for CIFAR10–LSUN-Resize and 5478.36 seconds for CIFAR100–LSUN-Resize. These measurements indicate that Medix filtering is feasible at this scale and does not impose large GPU memory overhead.

Table 6: Runtime and memory profile for Medix filtering on an NVIDIA A100 80GB GPU using approximately 15,000 unlabeled samples per InD-OOD pair.

InD-OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Our setting differs from semi-supervised open-set recognition (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in both target metric and use of unlabeled data. In OpenML Saito et al. (2021), for example, the goal is to improve a standard classifier despite outliers in SSL. More broadly, SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) are evaluated by whether classification remains strong despite outliers, whereas Medix treats the unlabeled pool as evidence for building a dedicated OOD detector, so the relevant comparison is detection quality rather than SSL classification accuracy.

The second distinction is how outliers are used. SSL methods generally suppress OOD samples as noise to protect classifier training, whereas Medix explicitly identifies candidate OOD samples and uses them as negative evidence for detector training. This changes the empirical question from whether outliers can be ignored to whether they can be extracted with low error and converted into improved OOD detection.

The techniques also differ. OpenMatch uses consistency regularization, enforcing similar classifier outputs across stochastic transformations to encourage smooth decision boundaries. Medix instead uses median-based gradient filtering, exploiting robustness to extreme values to identify candidate OOD samples. Its analysis explicitly bounds errors through contamination, concentration, and separation terms, which align with the empirical comparisons reported in the experiments.

B Broader Impacts and Limitations

Improved OOD detection has practical significance because deployment errors in healthcare, autonomous systems, finance, and other high-stakes domains can arise from inputs outside the training distribution. Medix contributes a median-centric mechanism for extracting OOD evidence from unlabeled wild data, improving the empirical basis for safer detector training while retaining theoretical error control. Future work could combine this filtering signal with generative outlier synthesis or adapt it to dynamic settings where OOD distributions change over time.

A limitation is that Medix assumes access to an unlabeled mixture containing both InD and OOD samples, and real deployment mixtures may be noisy, corrupted, or unevenly sampled. This assumption is not unique to Medix; it is the empirical condition that wild-data OOD detection methods must address. We make the assumption quantitatively explicit in the main theorems, including the geometric separation condition discussed in Remark 4.3. We also relax the tail requirement in Theorem C.3, which removes sub-Gaussianity and keeps a robustness guarantee under bounded second moments, albeit with weaker rates. Thus, the limitations are expressed as measurable conditions that can guide future extensions.

C Proof of Theorem

Table 7 summarizes the notation used to state the filtering errors, sample counts, gradient quantities, and concentration parameters in the theory. The table is intended to make the quantitative bounds easier to compare with the empirical filtering and detection metrics. Unless stated otherwise, bold symbols denote vectors, $\mathcal{X}$ denotes the input space, and all gradients are computed with respect to a fixed pretrained model f_ϕ .

Table 7: Notation used in the filtering-error bounds, including gradient summaries, sample counts, contamination terms, and concentration parameters.

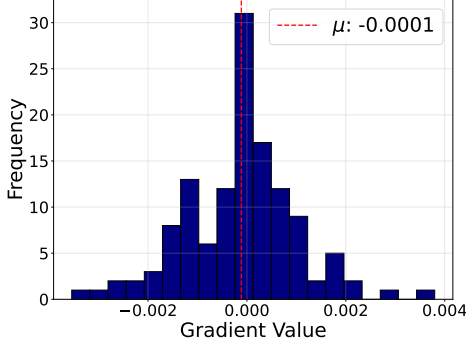
Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

C.1 Bound on the Inlier Misclassification Rate

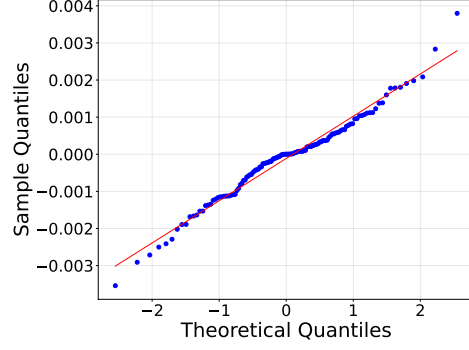
Theorem C.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Proof. The proof separates the inlier error bound into the same two quantities used in the theorem statement. In **Step 1**, we quantify the fraction η of InD samples whose gradients fall outside a high-probability ball around $\bar{\mathbf{V}}_{\text{in}}$; the remaining InD samples are the *good inliers*. In **Step 2**, a swapping argument for the EWM objective shows that excluding too many good inliers would produce a subset with a strictly smaller median-to-reference distance, contradicting optimality. Combining these pieces yields the stated upper bound on ERR_{in} .



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Illustration of InD sample gradients exhibiting sub-Gaussian behavior in each coordinate. (left) Histogram of gradient values (CIFAR-100 InD data) showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian distribution, confirming alignment with sub-Gaussianity.

Step 1: Concentration of InD Gradients. Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ) of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

The following norm comparison converts coordinate-wise concentration into an ℓ_2 deviation bound:

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

Combining (6) across $j = 1, \dots, d$ via a union bound gives the probability that an InD gradient falls outside the accepted radius:

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

The goal is to upper bound the empirical fraction of these bad inliers among the m_{in} InD points.

From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

704 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

705 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

706 Since X is a sum of bounded Bernoulli indicators, Hoeffding's inequality converts the expectation
707 bound into a high-probability fraction bound. For any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

708 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

709 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

710 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

711 Dividing by m_{in} gives the concentration contribution to the inlier error rate:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

712 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
713 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
714 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
715 called *good inliers*.

716 **Step 2: Contradiction via a Swapping Argument** We next show that an optimizer of the EWM
 717 objective cannot exclude too many good inliers. Suppose instead that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

718 Then one can replace far-away points with good inliers to construct $\hat{\mathcal{S}}$ whose EWM is strictly closer
 719 to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$, contradicting optimality.

720 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 721 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

722 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

723 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

724 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

725 This is the number of good inliers available to swap into $\mathcal{S}^*$ while preserving cardinality.

726 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 727 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

728 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

729 which measures the excluded-good-inlier fraction in the candidate minimizer. Rearranging gives:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

730 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

731 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

732 Because at most a η fraction of inliers lie outside B_ε and the OOD fraction is π , the total number of
 733 samples outside B_ε is bounded by:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

734 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

735 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_{\varepsilon}}{2}, \quad (33)$$

736 So $m'_{\varepsilon} = m_{\varepsilon}/2 + \eta m_{\text{in}}$, which implies

$$m_{\varepsilon} - m'_{\varepsilon} = \frac{m_{\varepsilon}}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_{\varepsilon} = m_{\text{in}}^* + \frac{m_{\varepsilon}}{2} + \eta m_{\text{in}}. \quad (35)$$

737 Since $\eta m_{\text{in}} > 0$, the swap gives points inside B_{ε} a strict majority:

$$m_{\text{in}}^* + m'_{\varepsilon} > m_{\varepsilon} - m'_{\varepsilon}, \quad (36)$$

738 so the points lying inside B_{ε} form a strict majority in $\hat{\mathcal{S}}$.

739 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 740 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

741 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 742 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 743 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

744 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

745 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 746 increase it, so the median distance from μ_j does not increase and possibly decreases.

747 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

748 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

749 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 750 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1 - \pi)}. \quad (41)$$

751 □

752 C.2 Bound on the Outlier Retention Rate

753 We next quantify the complementary filtering error: the fraction of OOD points that remain in
 754 the subset returned by the MEDIX filter. The bound parallels the inlier misclassification result in
 755 Appendix C.1, but the controlling comparison is now between the OOD gradient mean and the InD
 756 reference gradient. Thus, the rate makes explicit how vector separation, OOD concentration, sample
 757 size, and the InD-to-OD mixture ratio determine outlier retention.

758 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
 759 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

760 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

761 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

762 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

763 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

764 **Proof. Step 1: Concentration of OOD Gradients.**

765 The first term in the retention bound counts OOD gradients that look too close to the InD reference.
 766 Because each coordinate of $g(x)$ is sub-Gaussian and the OOD mean is separated from $\bar{\nabla}_{\text{in}}$ by at
 767 least $\Delta\sqrt{d}$, the probability that a true OOD point falls inside the InD-centered tolerance ball is at
 768 most $p_{\text{out}}(\varepsilon)$. Equivalently, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

769 has probability at most $p_{\text{out}}(\varepsilon)$.

770 **Counting the “ambiguous” OOD points.** We make this probability statement empirical by counting
 771 OOD examples whose gradients fall inside the same tolerance ball. For every OOD example x_i define
 772 the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

773 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent, so their sum directly measures the
 774 number of OOD samples that are statistically difficult to distinguish from InD gradients. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

775 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality converts the expected ambiguity
 776 count into a high-probability finite-sample bound. For any $s > 0$,

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

777 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

778 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

779 which holds with probability at least $1 - \delta$. On this event, at most an η_{out} fraction of the OOD pool
 780 can be close enough to the InD reference to be counted as ambiguous.

781 **Step 2: Contradiction via a Swapping Argument** The remaining terms arise from comparing any
 782 retained separated OOD mass against the number of InD points available for a better median-centered
 783 subset. Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

784 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_\varepsilon^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1 - \pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

785 Thus, after excluding the ambiguous OOD count controlled by η_{out} , more than half the points in $\mathcal{S}_{\text{in}}^*$
 786 are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

787 By the definition of the coordinate-wise median, this majority of separated OOD gradients forces the
 788 EWM of the retained subset to remain at least an $\varepsilon\sqrt{d}$ distance from the InD reference, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

789 Now compare this candidate subset with a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated
 790 OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist
 791 because the ambiguity count is below $m_{\text{out}}\eta_{\text{out}}$ and the available InD pool has size m_{in} .

792 After the swap, at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} -$
 793 $\varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, so the EWM is no farther than the same tolerance from the InD reference:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

794 Combining this improved median distance with (47) yields a strict comparison against the assumed
 795 optimum:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

796 This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$. Therefore, the assumed retention rate is too large. With
 797 probability at least $1 - \delta$, the outlier retention rate is bounded by the sum of the OOD tail probability,
 798 the finite-sample deviation term, and the mixture-ratio penalty:

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}, \quad (50)$$

799 □

800 C.3 Theorem C.3 without Sub-Gaussianity

801 **Theorem C.3** (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients*
 802 *of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment*
 803 *bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla\ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla\ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

804 Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and
 805 confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of
 806 the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq 2 \underbrace{\left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1 - \pi)}}_{\text{Contamination term}}.$$

807 **Differences with the original bound.** The sub-Gaussian version obtains a sharper concentration
 808 term,

$$2 \exp\left(-\frac{\varepsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

809 which decreases exponentially with dimension d . The finite-fourth-moment version replaces that
 810 term with

$$\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2}$$

811 which decays as $O(1/d)$. Thus the relaxed assumption permits heavier tails at the cost of a weaker
 812 rate. Both versions retain the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

813 which shrinks as m_{in} increases and is independent of the tail assumption.

814 *Proof.* The proof mirrors the sub-Gaussian argument but changes the quantitative concentration tool.
 815 First, we bound the fraction of “bad” InD samples, meaning those far from ∇_{in} , using only the finite
 816 fourth moment assumption. Second, we reuse the same swapping argument to convert that fraction
 817 into the inlier misclassification bound. The resulting structure preserves the contamination term while
 818 weakening the concentration rate.

819 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

820 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 821 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \varepsilon \sqrt{d}, \quad (52)$$

822 and “good” otherwise. The target quantity is the fraction of bad inliers among the m_{in} InD samples
 823 in S_{wild} .

824 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, the moment-based comparison proceeds through the sum of squared
 825 coordinate deviations. Assume that the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across
 826 $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

827 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

828 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

829 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

830 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

831 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

832 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

833 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

834 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
835 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}} p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

836 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
837 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

838 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

839 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

840 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

841 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
842 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

843 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
844 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1 - \pi)}, \quad (68)$$

845 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
846 thereby contradicting the optimality of $\mathcal{S}^*$.

847 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
848 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

849 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

850 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1 - \pi)}, \quad (70)$$

851 Define the swap size as

$$m'_\epsilon := \min \{ m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (71)$$

852 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
853 available for swapping in.

854 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
855 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

856 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

857 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

858 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1 - \pi)}, \quad (74)$$

859 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

860 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
861 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

862 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

863 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

864 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

865 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

866 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

867 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
868 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

869 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
870 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
871 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

872 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

873 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
874 increase it, so the median distance from μ_j does not increase and possibly decreases.

875 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

876 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

877 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
878 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

879 □